NETWORKED CONTROL
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EXERCIZES

1) Consider the following continuous-time system $\mathcal{S}$.

$$\begin{cases} \dot{x}(t) & = & 2u(t) \\ y(t) & = & 2x(t) \end{cases}$$

A) Assume that the system is controlled using a continuous-time control law of the type $u(t) = Ky(t)$. Define the range of values of K that make the closed-loop system asymptotically stable.

B) Compute the parameters $F, G,$ and H of the corresponding discrete-time dynamical equation

$$x_{k+1} = Fx_k + Gu_k, \qquad y_k = Hx_k$$

with sampling time $h = 1$ s.
Also, assuming that the system is controlled using a digital control law of the type $u_k = Ky_k$, define the range of values of K that make the closed-loop system asymptotically stable.

C) Assume that the system $\mathcal{S}$ is controlled through the networked control system (NCS) sketched in the following figure.

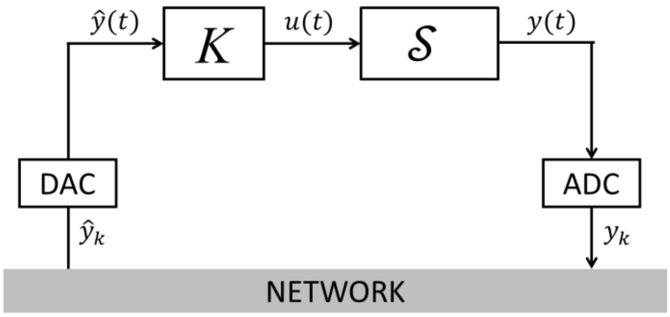

Figure 1: Networked control system

Answer the following questions considering the case where $h = 1$ s and $K = -\frac{1}{4}$.

a. Is the NCS asymptotically stable in case the delay is $\tau = 0$ s and in absence of packet dropouts?
b. Is the NCS asymptotically stable in case the delay is $\tau = 0.2$ s and in absence of packet dropouts?
c. Assume now that the NCS is subject to packet dropouts and in case the delay is $\tau = 0.2$ s. Variable θ_k has the following periodic behavior.

k	1	2	3	4	5	6	7	8	9	$\cdots$
θ_k	0	0	1	0	0	1	0	0	1	$\cdots$

Is the corresponding NCS asymptotically stable?

2) Draw a typical scheme of a **networked control system** (e.g., with remote controller), where the definition and the role of all its elements and involved signals (e.g., **data records** and **data images**) must be specified and properly commented. Comment on the role and function of **AD and DA converters and network interfaces**.

3) Consider a **networked control system** with an **ideal communication channel** (i.e., no delays or packet dropouts) with **bandwidth *B*** and **with signal-to-noise ratio *S/N***.

A) Define briefly the **channel capacity** R_{max}. How do you compute R_{max} ?

B) In case a **digital quantized controller** is used, where the scalar input variable u_k is transmitted over a communication channel with rate R and in packets of length N_B bits, and where the system eigenvalues are $\lambda_1, \dots, \lambda_n$, what is the **main tradeoff** to satisfy in the choice of N_B and of the sampling time h? Motivate adequately the response.

C) In case a digital quantized controller is used to control the system

$$\begin{cases} \dot{x}_1(t) = 2x_1(t) + x_2(t) + 2u(t) \\ \dot{x}_2(t) = 2x_2(t) + 3x_1(t) \\ y(t) = x_2(t) \end{cases}$$

and with sampling time $h = 0.2$ s, what is the minimum number of bits N_B encoding the control input required to make the system **boundable**?